

Phase 10A - phase10a_report

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PHASE 10A — DECISION EXPERIENCE, POWER BI (DATA MARTS + DASHBOARDS 1-7)

Real Estate AI Decision Intelligence Platform — Phase Completion Report

Phase 10 was split into 10A (Power BI architecture + all 7 dashboards) and 10B (the AI Explanation Layer, §10.10), per explicit direction. This report covers 10A.

1. PHASE STATUS

PHASE 10A — COMPLETE. A governed data-mart layer and all 7 dashboards (§10.3-10.9) are built. One real data bug was found during visual QA and fixed before delivery.

2. OBJECTIVE

Deliver the validated decision intelligence from Phases 2-9 to business users (§10.1) through a Power BI-ready data architecture and all 7 required dashboards, without recomputing any governed number in the presentation layer.

3. WHY HTML MOCKUPS INSTEAD OF A .PBIX FILE

Stated plainly rather than glossed over: this build environment cannot run Power BI Desktop and cannot produce a genuine `.pbix` binary. What it *can* do, and what this phase actually delivers, is the two things that determine whether a real Power BI build will be correct: (a) clean, governed data marts, and (b) working, data-accurate previews of every required dashboard. `docs/power_bi_architecture.md` includes a literal build guide for turning these marts into a real `.pbix`, plus a DAX measures library written for that purpose (not executed here, since no `.pbix` exists in this environment).

4. WHAT WAS BUILT

Data mart layer (`src/marts/build_marts.py` → `data_marts/*.csv` , 12 files): reads exclusively from Phase 6/7/8/9's already-governed CSV outputs, plus one direct read of `core.market_monthly` for locality market intelligence (reusing Phase 8's own tested `market_intelligence()` function, since Phase 2 produced SQL views but no CSV export). No financial, risk, geo, or decision math is recomputed anywhere in this layer — verified by inspection: `build_marts.py` contains zero IRR/NPV/VaR/location-score formulas.

7 dashboards (`dashboards/*.html`), each a self-contained, data-accurate HTML page (Chart.js + embedded real data), matching a consistent design system (`src/dashboards/shell.py`):

Dashboard	Master Prompt Section	Source marts
1 — Executive Overview	§10.3	mart_01, mart_02, mart_06
2 — Market & Locality Intelligence	§10.4	mart_02
3 — Property & Investment Analysis	§10.5	mart_03
4 — Financial & Risk Intelligence	§10.6	mart_04, 04b, 04c
5 — Geo Intelligence	§10.7	mart_05
6 — Decision Intelligence	§10.8	mart_06, 06b, 06d
7 — Data Quality & Monitoring	§10.9	mart_07, mart_06c

`power_bi_architecture.md` : star-schema data model, DAX measures library, `.pbix` build guide, governance-preservation rules, refresh strategy.

5. A REAL BUG FOUND DURING VISUAL QA

Rendering Dashboard 3 to check the layout (`wkhtmltoimage` , since no live browser is available in this sandbox either) surfaced a genuine data defect: **property_id=1's locality showed as "nan"** in the comparables table. Root cause: `build_property_investment_mart()` originally sourced `locality_name` via a merge against Phase 6's financial summary — which only covers the 25 properties **with** full financial data. Property 1 was deliberately included in this mart specifically to demonstrate Phase 9's **INSUFFICIENT** decision path (it has no expense/rental history), so it has no Phase 6 row, and the merge silently left its locality blank.

Fix: `locality_name` is now sourced independently, directly from `properties.csv` → `projects.csv` → `localities.csv` , for every property regardless of its financial-data completeness. Verified after the fix: zero NaN locality names across all 26 properties in the mart. This is the same class of finding every prior phase has surfaced — a real join gap, caught by actually looking at the rendered output rather than trusting the code to be correct because it ran without an error.

5B. POST-DELIVERY AUDIT FIX — inherited `roi_total_holding_period` bug (now fixed), and a missing test suite added

An independent review (the same review that found this same inherited-stale-copy pattern in Phases 7 and 9) found two gaps:

1. `src/finance/engine.py` was pulled into this phase *before* Phase 6's own post-delivery audit fixed `roi_total_holding_period` (it was computing a gross cash multiple instead of a net return, overstating ROI by exactly 100 percentage points in every case). The bug had propagated further than in Phase 7/9: `sources/phase6/03_financial_metrics.csv` , `04_scenario_comparison.csv` , `05_sensitivity_analysis.csv` , and `sources/phase7/05_extended_scenario_comparison.csv` — **the static, pre-computed CSVs this phase's mart builder reads — all still carried the pre-fix values**, so `mart_04_financial_risk.csv` and `mart_04c_scenarios_extended4.csv` (the actual Power BI-

facing deliverable) were wrong at the data layer, even though no dashboard HTML visibly renders `roi_total_holding_period` as a labeled number (checked directly: only `cap_rate`, `irr`, `npv`, `dscr`, `probability_negative_npv`, and `var_95_npv` are rendered in Dashboard 4's table/charts).

2. **No automated test suite existed for this phase** (`src/finance/test_phase6.py` and `src/geo/test_phase8.py` were carried over from their source phases, but nothing tested the marts or dashboards this phase actually built).

Fixed: re-copied Phase 6's corrected `engine.py`; replaced the four stale source CSVs with their corrected equivalents (property_id sets verified identical via diff before substituting, so no row was added/dropped, only the ROI column corrected); reran `build_marts.py` (all 12 marts) and all 7 dashboard-generation scripts. Property 2731's Base-case ROI in `mart_04_financial_risk.csv`: **223.8% → 123.8%**, consistent with its 17.85% IRR compounded over the 5-year hold. Portfolio KPIs (`mart_01`) and the decision distribution (INVEST 6 / HOLD 12 / AVOID 7 / INSUFFICIENT 1) were independently re-derived from the regenerated marts and came back identical to what this report already claimed – confirming no decision-relevant figure was ever affected by this bug.

Added: `src/marts/test_phase10a.py`, 9 new tests – an ROI net-return regression check, a cross-scenario IRR/ROI monotonicity check, the property_id=1 NaN-locality-name regression check (locking in Section 5's fix), portfolio-KPI-vs-Phase-9-source consistency, flood-risk-never-invented, a structural check that `build_marts.py` contains no recomputed financial/risk formulas (enforcing Section 10.2 directly, not just by inspection), and dashboard-file-existence checks. **All 9 passing.**

6. HONEST FIGURES CARRIED INTO THE DASHBOARDS (NOT SOFTENED)

- Dashboard 6's mean decision confidence: **77.3/100**, with a visible note that the practical ceiling is ~82/100 because Phase 7's risk simulation is permanently UNCALIBRATED — not smoothed over or hidden behind a “looks good” green KPI card.
- Dashboard 5: **90 of 90 localities show INSUFFICIENT DATA for flood/environmental risk** — not filled with a plausible-looking estimate.
- Dashboard 6's top-risk table surfaces “Flood/environmental risk data is not available for this locality” as the single most common flagged risk across the demo batch (26 occurrences) — the platform's own data gap is visible in its own risk-reporting dashboard, which is the correct behavior, not an embarrassment to hide.
- Dashboard 1's Alerts panel is a direct carry-forward of each phase's own build-time findings (the Phase 3 locality-price bug, Phase 7's appreciation-dominance finding, etc.) — not regenerated or reworded here.

7. MASTER PROMPT REQUIREMENTS COVERAGE

Section	Requirement	Status
10.2	Power BI consumes governed outputs, no DAX duplication	✓ <code>build_marts.py</code> , zero recomputed math
10.3	Dashboard 1 — Executive Overview	✓
10.4	Dashboard 2 — Market & Locality Intelligence	✓
10.5	Dashboard 3 — Property & Investment Analysis	✓
10.6	Dashboard 4 — Financial & Risk Intelligence	✓
10.7	Dashboard 5 — Geo Intelligence	✓
10.8	Dashboard 6 — Decision Intelligence	✓
10.9	Dashboard 7 — Data Quality & Monitoring	✓

12	Testing at every layer	✓ <code>src/marts/test_phase10a.py</code> , 9/9 passing (added during post-delivery audit – see Section 5B)
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8. KNOWN LIMITATIONS (see also `docs/limitations.md`)

1. No genuine `.pbix` file — HTML mockups + data marts + a build guide, honestly stated as such (Section 3).
2. All 7 dashboards are built on the same `n=26` demo batch every prior phase used (25 properties with full financial data + property 1 for the INSUFFICIENT case) — not the full 8,000-property portfolio.
3. Dashboard interactivity (Chart.js, Google Fonts) requires internet access to render correctly — this sandbox’s own `bash` tool could not fetch those CDN resources to screenshot-verify final rendering, though the HTML/CSS structure was confirmed correct via a degraded-mode render.
4. The DAX measures library (`power_bi_architecture.md` Section 4) is documentation, not executed or tested code — no Power BI runtime exists in this environment to validate DAX syntax.
5. Refresh automation (Section 7 of the architecture doc) is a documented target only, not implemented.

9. DEPENDENCIES

Phases 2 (semantically, via `market_monthly`), 4, 6, 7, 8, 9 — all consumed as already-governed outputs, none recomputed.

10. SECURITY CONSIDERATIONS

No credentials in this phase (CSV-only mart building). The architecture doc’s Section 5 build guide notes where live PostgreSQL credentials would be configured in an actual Power BI Desktop session — outside this deliverable’s scope.

11. PERFORMANCE CONSIDERATIONS

Mart build (`build_marts.py`) completes in a few seconds — it’s flat CSV joins over already-small governed outputs (max 8,000 rows for the properties table, everything else under 500 rows). Not a performance-sensitive step.

12. PRODUCTION READINESS

Per Section 24: **Production-Oriented Prototype**, and explicitly *not* “Power BI dashboards are live” — they are HTML mockups plus a complete, followable build guide. A Power BI developer following `docs/power_bi_architecture.md` Section 5 can produce the real `.pbix` in well under a day, since the data model, measures, and exact visual/layout reference already exist.

13. NEXT PHASE DEPENDENCIES

Phase 10B (AI Explanation Layer, §10.10) should read `DecisionResult` objects from Phase 9’s orchestrator directly — the same source `mart_06*` already reads from — and its output should be designed to slot into Dashboard 6 as an additional panel, without requiring any change to this phase’s data marts.

14. FINAL STATUS

PHASE 10A: COMPLETE. Governed data-mart layer (12 CSVs) and all 7 required dashboards built, with one real data bug found during visual QA and fixed before delivery. Power BI architecture, DAX measures library, and a .pbix build guide are documented. Ready for Phase 10B (AI Explanation Layer).